

# Empirical Proof Record: VALIDATED

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**Proof ID:** 89e1ebd7bf01f63db9c1a8d34538cf9b...

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### 1. Claim

Across 30 synthetic three-group datasets of size 30 per group sweeping effect size from null to strong (seed=20260518), “`scipy.stats.f_oneway`“, “`statsmodels.stats.anova.anova_lm`“ (via OLS), and “`pingouin.anova`“ produce F-statistics AND two-sided p-values that agree to  $\leq 1e-09$  across all pairwise comparisons. (RFC 0002 Phase E1: cross-framework validation; three-way ANOVA variant.)

- **Operationalisation:** Generate N three-group datasets where each group has size n and is drawn from  $\text{Normal}(\mu_g, \sigma^2)$  for a sweep of  $(\mu_0, \mu_1, \mu_2)$  configurations including null (all  $\mu$  equal) and alternatives. Compute F and two-sided p under `scipy.stats.f_oneway`, `statsmodels OLS+anova_lm`, and `pingouin.anova`. Compute `max_abs_diff` = max over (dataset, statistic  $\in \{F, p\}$ , backend-pair) of  $|\text{stat}_a - \text{stat}_b|$ . VALIDATED iff `max_abs_diff`  $\leq$  tolerance.
- **Threshold:** `max_absolute_anova_difference`  $\leq 1e-09$  F\_or\_p
- **H0:** At least one backend pair disagrees on the ANOVA F statistic OR the two-sided p value by more than the floating-point tolerance.
- **H1:** All three backends compute identical F AND p to floating-point tolerance across every dataset.

### 2. Verdict

**VALIDATED** — observed `7.105427357601002e-14`

30 datasets  $\times$  3 backends  $\times$  2 statistics; max pairwise  $|\Delta| = 7.105e-14$  on (F, `scipy_vs_statsmodels`) ( $\leq 1e-09$  tol)

### 3. Evidence

| Pillar                                 | Statistic                     | Value               |
|----------------------------------------|-------------------------------|---------------------|
| anova_scipy_vs_statsmodels_vs_pingouin | max_absolute_anova_difference | 7.105427357601002e- |

### 4. Pre-registration

- Registered at: 2026-05-23T23:09:39.603552+00:00
- Config hash: c8e368fe8b61be3829c27bb61d9b3208...
- Data hash: c8e368fe8b61be3829c27bb61d9b3208...

### 5. Signature

003ddcb88698d585e6b8623df154afb116598f57fa164010a75891167a8b0530